

19-200-0105B

Basic Mechanical Engineering

Dr. Harikrishnan S
Assistant Professor
Division of Mechanical Engineering

Syllabus

19-200-0105B/19-200-0205A BASIC MECHANICAL ENGINEERING

Course Outcomes:

On completion of this course the student will be able to:

1. Understand basics of thermodynamics and working of steam turbines
2. Understand basics of internal combustion engines, refrigeration and air conditioning
3. Gain knowledge on the working of hydraulic turbines and centrifugal pumps
4. Identify manufacturing methods encountered in engineering practice and understand mechanism of power transmission

Module I

Thermodynamics: Thermodynamics systems – open, closed and isolated systems, equilibrium state of a system, property and state, process, cycle, Zeroth law of thermodynamics- concept of temperature, temperature scales. First law – internal energy, enthalpy, work and heat, Different processes (isobaric, isochoric, isothermal, adiabatic and polytropic processes). Second law – Kelvin-plank and Clausius statements and their equivalence, Carnot Cycle (Elementary problems only). Thermodynamic properties of Steam, Steam Generator. Different types of boilers, boiler mountings and accessories. Formation of steam at constant pressure, working of steam turbines, compounding of turbines.

Module II

Internal Combustion Engines: Air standard cycles – Otto and Diesel cycles, working of two stroke and four stroke Petrol and Diesel engines, Carburetted and MPFI engines, fuel pump, fuel injector, ignition system, cooling system, lubricating system.

Refrigeration & Air-conditioning: Introduction to refrigeration and air -conditioning, Rating of refrigeration machines, Coefficient of performance, Simple refrigeration vapour compression cycle (Elementary problems only), Summer and winter air conditioning.

Module III

Hydraulic Turbines & Pumps: Introduction, Classification, Construction details and working of Pelton Wheel, Francis and Kaplan turbines, Specific speed (Definition and significance only),



Classification of water pumps, working of centrifugal pumps and reciprocating pumps (Theory of working principles only)

Power plants: Hydro-electric power plants, Thermal power plants, Nuclear power plants, Diesel power plants, Wind mills, solar energy (Working principles using schematic representations only)

Module IV

Introduction to Manufacturing Systems: Welding- different types of welding, resistance welding, arc welding, gas welding, Brazing and soldering, Different welding defects. Casting- different casting processes, sand casting, casting defects, Rolling- hot rolling and cold rolling, two high, three high, cluster rolling mills, wire drawing, forging, extrusion, Heat treatment of steel, elementary ideas of annealing, hardening, normalizing, surface hardening.

Power Transmission Methods and Devices: Introduction to Power transmission, Belt, Rope, Chain and Gear drive. Length of belt open and crossed. Ratio of belt tensions (Elementary problems only). Different types of gears (Elementary ideas only). Types and functioning of clutches.

References

1. Nag, P.K. Engineering thermodynamics. (Fifth Edition). McGraw Hill Education (India) Pvt. Ltd, New Delhi.(2013).
2. Gill, J.H. Smith Jr. and Ziurys, E.J. Fundamentals of internal combustion engines, Oxford & IBH, New Delhi.(1959)
3. Stoecker, W. F. Refrigeration and air conditioning. Tata McGraw Hill, New Delhi. (1980).
4. JagadishLal. Hydraulic machines. Metropolitan Book co, New Delhi.(1994)
5. Raghavan, V. Material science and engineering, Prentice Hall of India, New Delhi. (2004)
6. Rajendar Singh.Introduction to basic manufacturing processes and workshop technology, New Age International, New Delhi. (2006).

Module-2b: Refrigeration and Air-Conditioning



Refrigeration and Air-Conditioning

- **Refrigeration**:- Transfer of heat from lower temp region to higher temperature region
- Heat always flow from higher temp to lower temp
- Working fluid Refrigerant
- Refrigerator:- Maintain refrigerated space at a low temperature by removing heat from it. Discharging this heat to a higher temp medium is merely a necessary part of the operation, not purpose
- Heat Pump:- To maintain space at a high temp. This is accomplished by absorbing heat from a low temperature source, and supplying this heat to a warmer medium

Refrigeration and Air-Conditioning

- **Air-conditioning**:- Defined as the treatment of indoor air in order to control certain conditions required for human comfort.
- The desirable conditions may be
 - Temperature
 - Humidity
 - Dust particle level
 - Odor level, and air motion
- Heat always flow from higher temp to lower temp

Applications



Fun Facts! :- Air Curtains

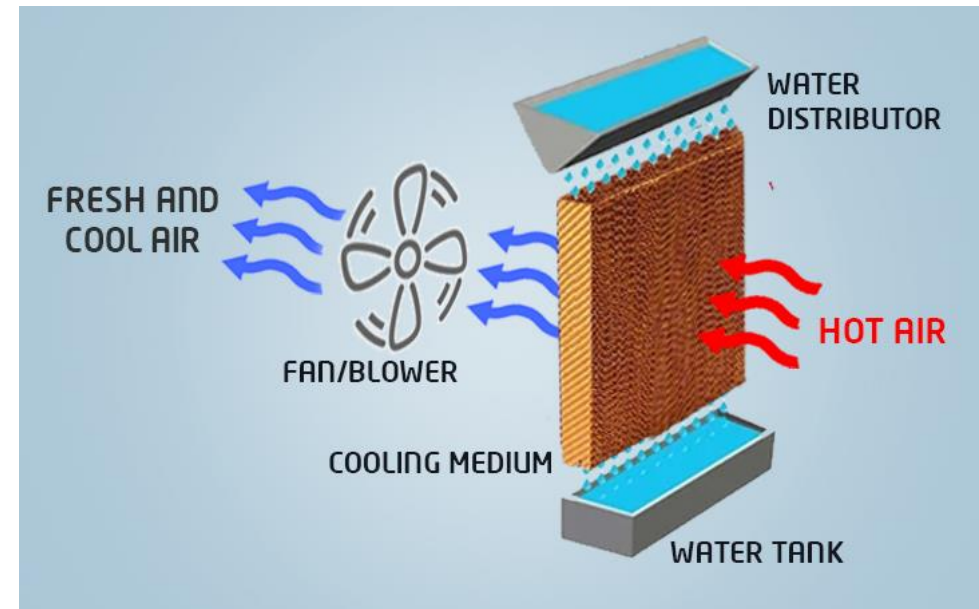


Air Curtains

- It keeps the atmosphere clean from pest and insects, dust, airborne, pollution, smells, odours and stops draughts and cold or hot air entrance
- It keeps the atmosphere clean from pest and insects, dust, airborne, pollution, smells, odours and stops draughts and cold or hot air entrance



Air coolers



Sweating



Drying cloths



Refrigerator and Heat Pump

- Heat:- high temperature to low temperature
- Transfer of heat from low temperature to high temperature:- device is known as Refrigerator
- Device that transfers heat from a low-temperature medium to a high-temperature one is the heat pump:- Heat pump
- Refrigerators and heat pumps operate on the same cycle but differ in their objectives.

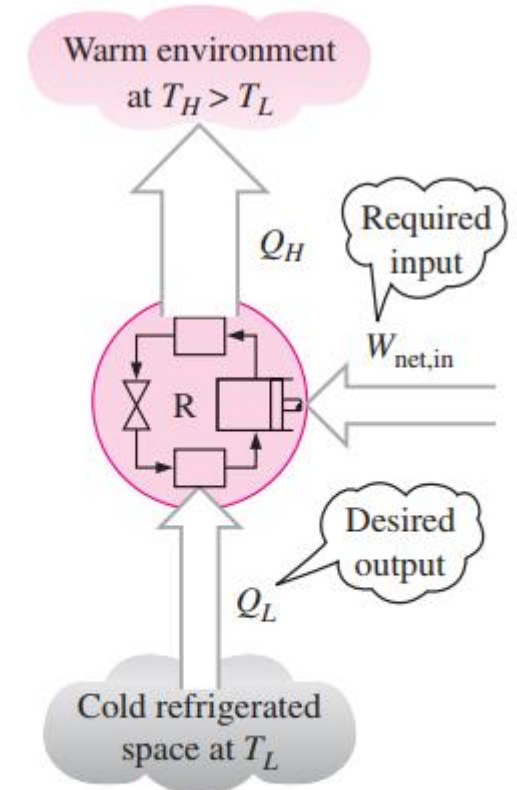
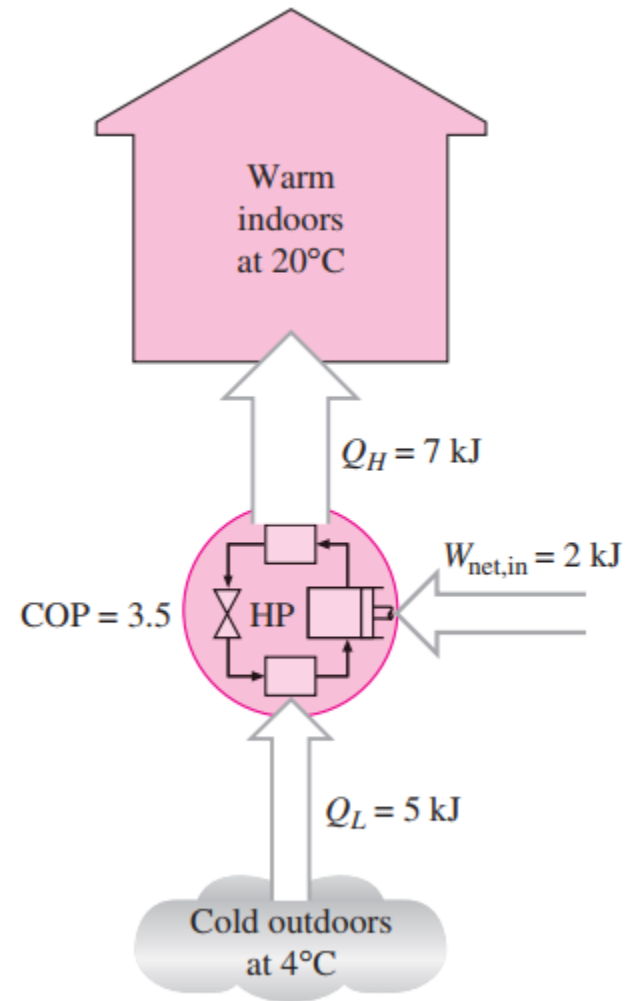


FIGURE 6-20

The objective of a refrigerator is to remove Q_L from the cooled space.

Heat Pump



Coefficient of Performance

- The objective of a refrigerator is to maintain the refrigerated space at a low temperature by removing heat from it. Discharging this heat to a higher-temperature medium is merely a necessary part of the operation, not the purpose.
- The efficiency of a **refrigerator/heat pump** is expressed in terms of the coefficient of performance (COP)

$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{W_{\text{net,in}}}$$

$$\text{COP}_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{Q_H/Q_L - 1}$$

$$\text{COP}_{\text{HP}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_H}{W_{\text{net,in}}}$$

$$\text{COP}_{\text{HP}} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

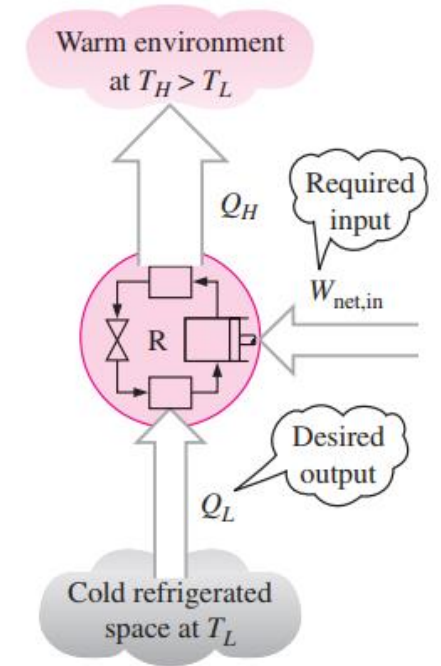


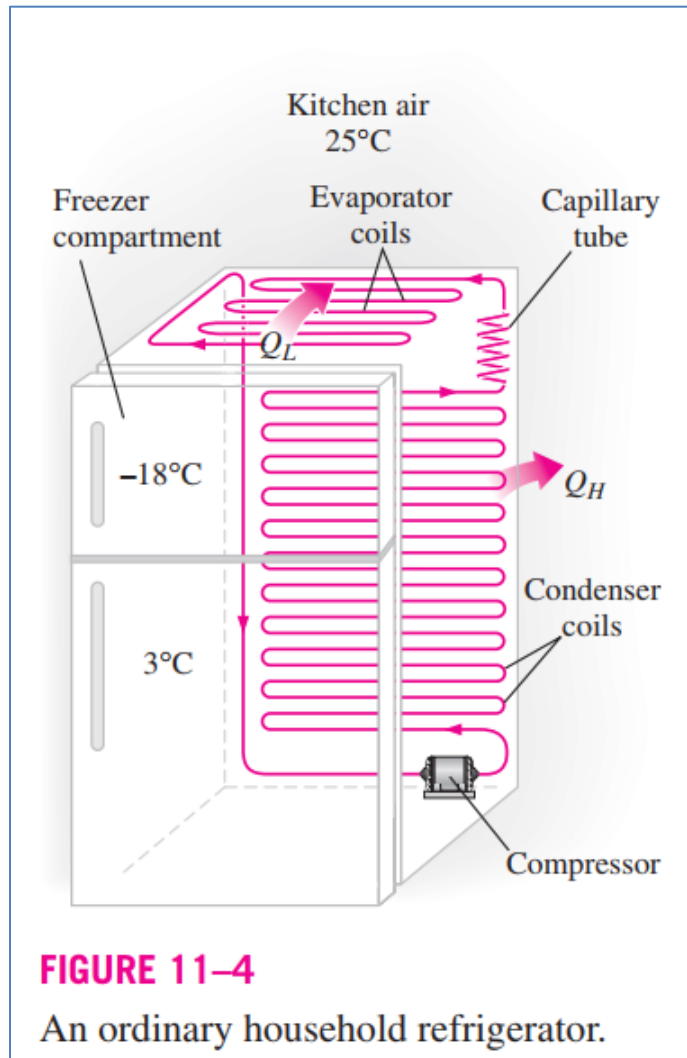
FIGURE 6-20

The objective of a refrigerator is to remove Q_L from the cooled space.

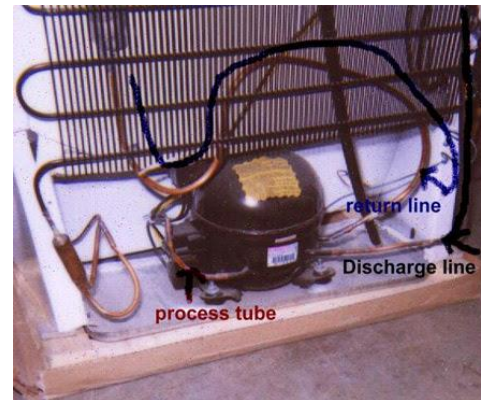
Unit of Refrigeration Capacity

- Ton of Refrigeration
- Rate of heat removal from refrigerated space is often expressed in terms of 'Ton of refrigeration'
- "Capacity of refrigeration system that can freeze 1 ton (2000 lb) of liquid water at 0°C into ice at 0°C in 24 Hours'
- Calculations
- 1Ton = 907kg, Latent heat of water/ice = 333.5 kJ/kg
- Heat extracted?
 - In kW
 - In kJ/Hr

Vapor Compression Refrigeration System (VCRS)



Capillary tube/Throttle valve



Compressor



Condenser

Vapor Compression Refrigeration System (VCRS)

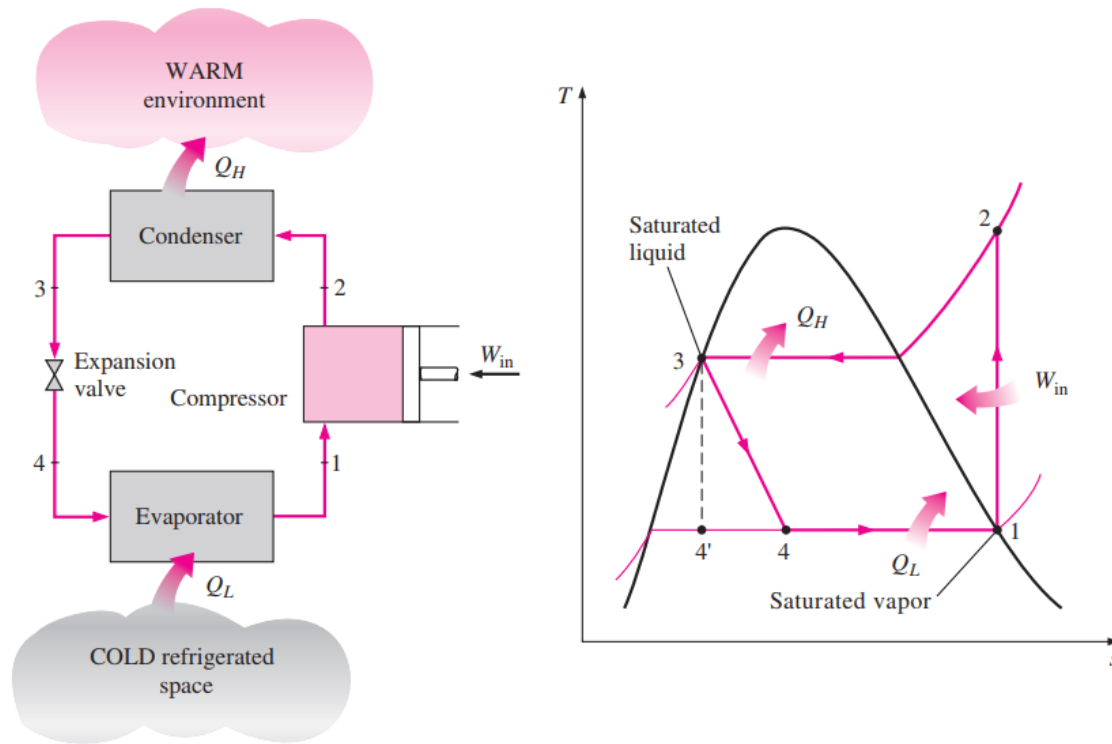


FIGURE 11-3

Schematic and $T-s$ diagram for the ideal vapor-compression refrigeration cycle.

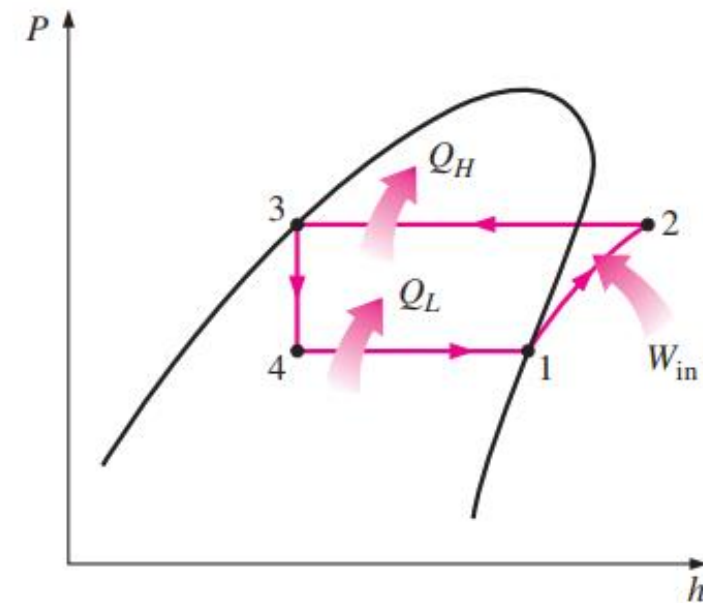


FIGURE 11-5

The $P-h$ diagram of an ideal vapor-compression refrigeration cycle.

Vapor Compression Refrigeration System (VCRS)

Processes

- 1-2: Isentropic compression
- 2-3: Isobaric heat rejection
- 3-4: Isenthalpic expansion
- 4-1: Isobaric heat addition

$$\dot{Q}_{TR} = 211 \text{ kJ/Hr} = 3.5 \text{ kW}$$

Air-Conditioning



Psychrometry



- Psychrometry is an engineering science that deals with the behaviour of moist air(dry air + water vapour mixture)
- The amount of water vapour in air plays important role in both comfort and industrial air conditioning

Fun Facts!: Silica Gel packets in packaging



Terms in Psychrometry

- **Dry air:** Air contains no water vapour
- **Moist air:** Mixture Of dry air and water vapour
- **Saturated air:** Air Which contains maximum amount of water vapour which air can hold at a given temperature and pressure
- **Dry bulb temperature:** Temperature of air measured by ordinary thermometer
- **Wet bulb temperature:** Temperature Recorded by a thermometer, when its bulb is covered by a wet cloth
- **Specific humidity:** Ratio of the mass of water vapour to the mass of dry air

Terms in Psychrometry



- **Specific humidity:** Ratio of the mass of water vapour to the mass of dry air
- **Relative humidity:** Ratio of mass of water vapour in a given volume of moist air at a given temperature to the mass of water vapour contained in the same volume of moist air at the same temperature, when the air is saturated

Psychrometric Processes

1. **Sensible Heating(C):** Increase the dry bulb temperature of air without change in its specific humidity(moisture content), using heating coil
2. **Sensible Cooling(G):** Decreases dry bulb temperature of air without change in its specific humidity(moisture content), using cooling coil
3. **Humidification(A):** Increase the specific humidity without change in dry bulb temperature, using humidifier
4. **Dehumidification(E):** Decreases the specific humidity without change in dry bulb temperature, using dehumidifier
5. **Heating with Humidification(B):** Increases both dry bulb temperature and specific humidity of air, using high temperature steam
6. **Heating with Dehumidification(D):** Increases the dry bulb temperature and decreases the specific humidity of air, using silica gel or alumina
7. **Cooling with Humidification(H):** Decreases the dry bulb temperature and increases the specific humidity of air, using cold water
8. **Cooling with Dehumidification(F):** Decreases both dry bulb temperature and specific humidity of air, using cooling coil with very low temperature

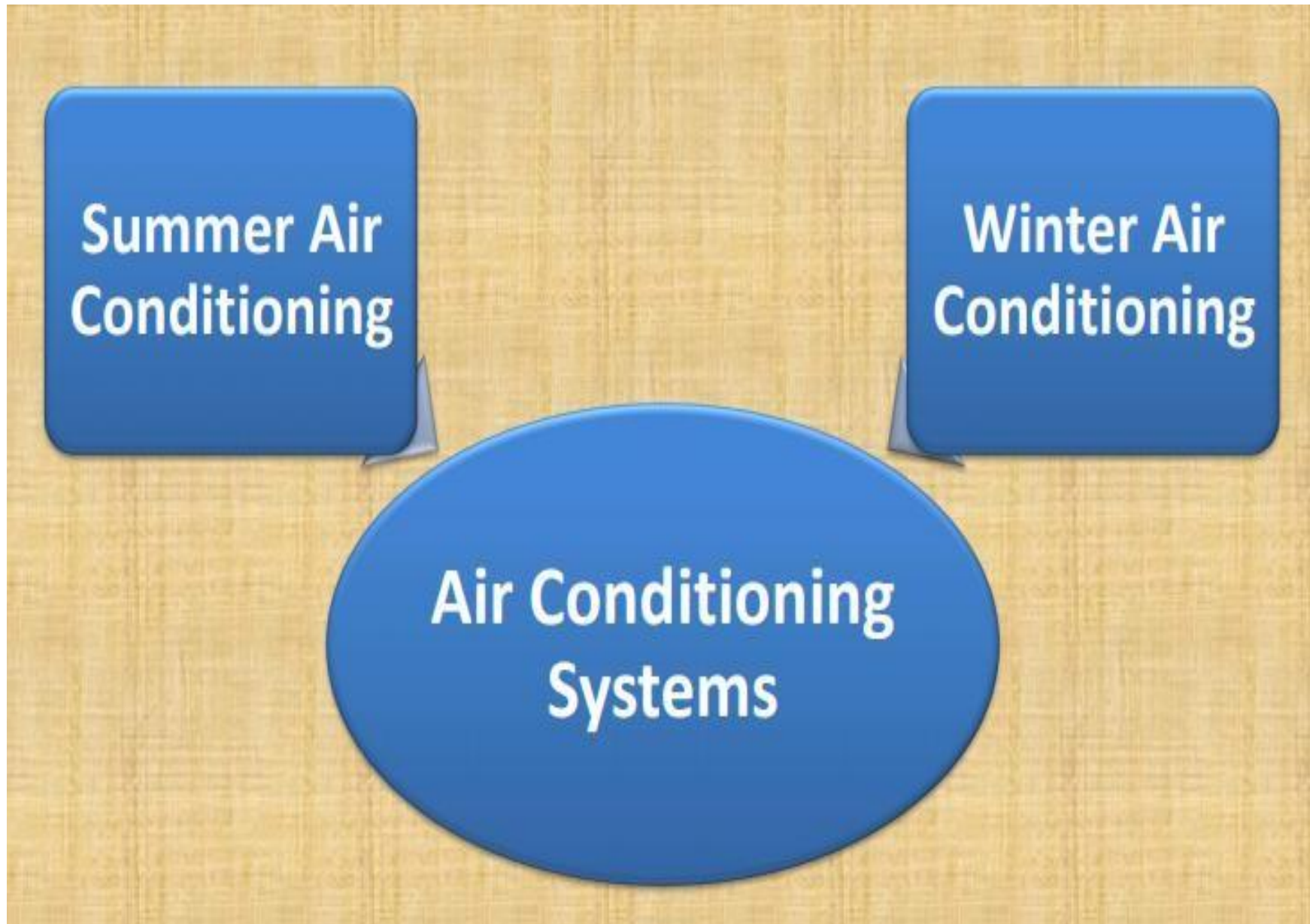
Human Comfort Conditions

- **Temperature:** 22°C - 27°C
- **Relative Humidity:** 55-65%
- **Air Velocity:** 0.2-0.5 m/s

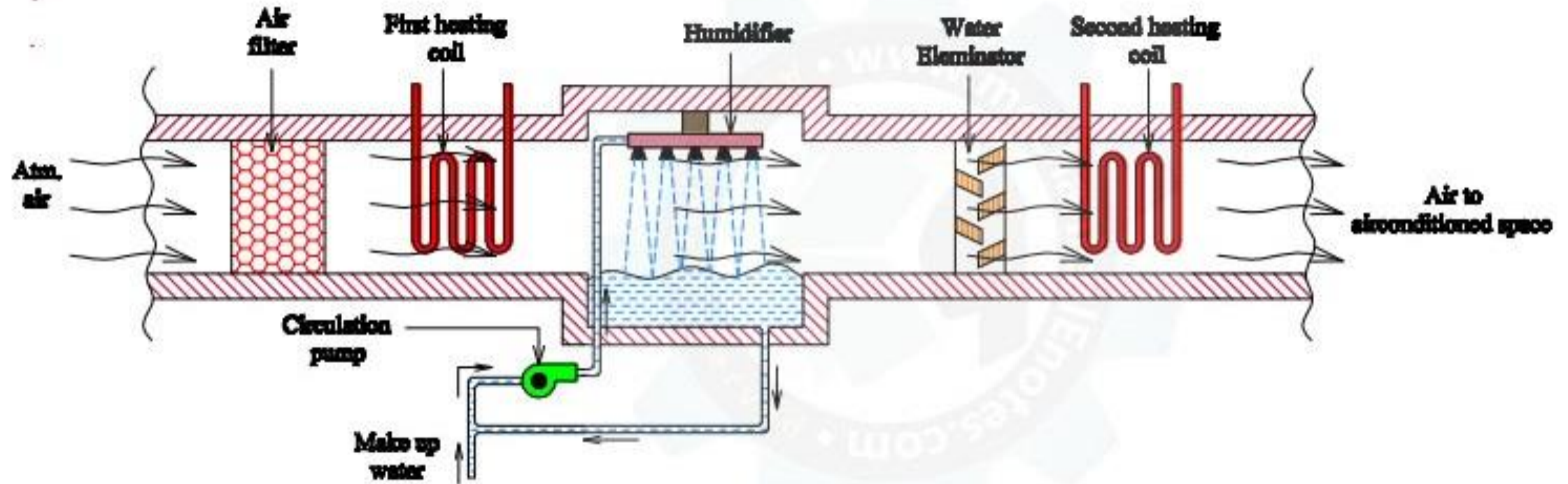
**Summer Air
Conditioning**

**Winter Air
Conditioning**

**Air Conditioning
Systems**



Winter Air-Conditioning



Winter Air-Conditioning

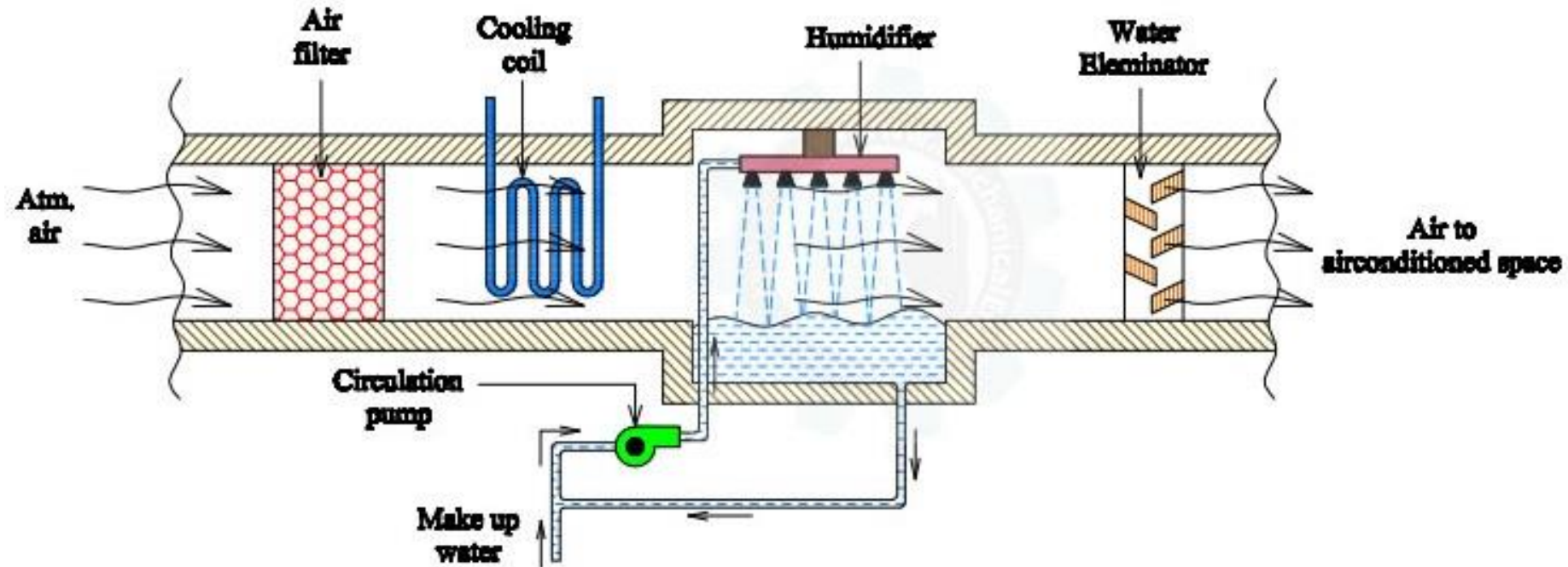
- Air is heated and generally humidified
- Outside air flows through the damper, and mixes up with recirculated air(obtained from the conditioned space).
- The mixed air passes through a filter to remove dust, dirt and other impurities
- Air now passes through a preheat coil in order to prevent the possible freezing of water and to control the evaporation of water in the humidifier
- The preheated air passes through a reheat coil to bring the air to the designed room temperature

Winter Air-Conditioning



- The humidification of air can be achieved in several ways, e.g. by bringing the air in contact with a wetted surface, or with droplets of water as in an air washer, by adding water droplets directly to air or by direct addition of dry saturated or superheated steam
- The most common method of humidifying air for these applications is by direct addition of dry steam to air

Summer Air-Conditioning



Summer Air-Conditioning

- Summer air conditioning system requires a refrigeration system that reduces the temperature of the air to temperatures much lower than the surroundings.
- Air is cooled and generally dehumidified
- Outside air flows through the damper, and mixes up with recirculated air (obtained from the conditioned space).
- The mixed air passes through a filter to remove dust, dirt and other impurities
- Air now passes through a cooling coil. The coil has a temperature much below the required dry bulb temperature of the air in the conditioned space
- The cooled air passes through a perforated membrane and loses its moisture in the condensed form which is collected in a sump
- Then air is heated by using a heating coil to bring the air to the designed room temperature

Thank You